

Homework 2
MSBA 400: Statistical Foundations for Data Analytics
Professor Rossi

Question 1

Q1, part A

In the class notes, we introduced the concept of R^2 . Show that the formula $R^2 = \frac{SSR}{SST}$ implies that R^2 is the square of the sample correlation coefficient between X and Y , r_{XY} . Hint: recall from the notes how the fitted regression line can be expressed in terms of deviations from the mean.

Q1, part B

In the class notes, we used intuition to argue the regression property, $\text{corr}(X, e) = 0$. Show this directly results from the formula for b_1 . Hint: substitute, $e_i = Y_i - b_0 - b_1 X_i$ into $\text{corr}(X, e)$.

Question 2 : More on Nearest Neighbor Approaches

The simplest view of nearest neighbor methods is to “slice” into the data for only a small interval of X values. This distribution is called the conditional distribution of Y given X .

Q 2, part A

Display a histogram of the conditional distribution of `emv` given that `luxury` is in the interval `(.2, .3)` in the `cars` dataset (from problem set 1).

Q 2, part B

Compute the mean of the conditional distribution in part A and compute a prediction interval that takes up 95% of the data (an interval that stretches from the .025 quantile (2.5 percentile) to the .975 (97.5 percentile)). Use the `quantile()` command.

Q 2, part C

Repeat part A and B for a much higher level of `luxury`, namely the interval `(.7,.8)`. Describe the difference between these two conditional distributions.

Q 2, part D

Explain why the results of part B and C show that `luxury` is probably (by itself) not sufficiently informative to give highly accurate predictions of `emv`.

Question 3 : Optimal Pricing and Elasticities

Q 3, part A

Use the `detergent` dataset to determine the price elasticity of demand for 128 oz Tide. Compute the 90 percent confidence interval for this elasticity.

Q 3, part B

One simple rule of pricing is the “inverse elasticity” rule that the optimal gross margin should be equal to the reciprocal of the absolute value of the price elasticity, i.e. $\text{Gross Margin} = \frac{1}{|\text{elasticity}|}$. For example, suppose we estimate that the price elasticity is -2 (a 1 per cent increase in price will reduce sales (in units) by 2 per cent. Then the optimal gross margin is 50 percent.

Suppose this retailer is earning a 25 per cent gross margin on 128 oz Tide. Perform appropriate hypothesis test to check if the retailer is pricing optimally at the 90 per cent confidence level?

Hints:

- i. use the inverse elasticity rule to determine what elasticity is consistent with a 25 per cent gross margin.
- ii. Use the confidence interval!

Questions 4-5 explore the sampling properties of least squares

Question 4

- a. Write your own function in R (using `function()`) to simulate from a simple regression model. This function should accept as inputs: b_0 (intercept), b_1 (slope), X (a vector of values), and σ (error standard deviation). You will need to use `rnorm()` to simulate errors from the normal distribution. The function should return a vector of Y values.
- b. Simulate Y values from your function and make a scatterplot of X versus simulated Y . When simulating, use the `vwretd` data from the `marketRf` dataset as the X vector, and choose $b_0 = 1$, $b_1 = 20$, and $\sigma = 1$. Then add the fitted regression line to the plot as well as the true conditional mean line (the function `abline()` may be helpful).

Question 5

Assume $Y = \beta_0 + \beta_1 X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Let $\beta_0 = 2$, $\beta_1 = 0.6$, and $\sigma^2 = 2$. You can make X whatever you like, for example you could simulate X from a uniform distribution

- a. Use your R function from question 4 to simulate the sampling distribution of the slope. Use a sample size of $N = 300$ and calculate b_0 & b_1 for 10,000 samples. Plot a histogram of the sampling distribution of b_0 .
- b. Calculate the empirical value for $\mathbb{E}[b_1]$ from your simulation and provide the theoretical value for $\mathbb{E}[b_1]$. Compare the simulated and theoretical values.
- c. Calculate the empirical value for $\text{Var}(b_1)$ from your simulation and provide the theoretical value for $\text{Var}(b_1)$. Compare the simulated and theoretical values.

Question 6

Standard errors and p-values.

- a. What is a standard error (of a sample statistic or an estimator)? How is a standard error different from a standard deviation?
- b. What is sampling error? How does the standard error capture sampling error?
- c. Your friend Steven is working as a data analyst and comes to you with some output from some statistical method that you've never heard of. Steven tells you that the output has both parameter estimates and standard errors. He then asks, "how do I interpret and use the standard errors?" What do you say to Steven to help him even though you don't know what model is involved?
- d. Your friend Xingua works with Steven. She also needs help with her statistical model. Her output reports a test statistic and the p-value. Xingua has a Null Hypothesis and a significance level in mind, but she asks "how do I interpret and use this output?" What do you say to Xingua to help her even though you don't know what model is involved?